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1. Quick Sort with all Cases

CODE:

```
//ch.sc.u4cse24110
//quick sort with all cases
#include <stdio.h>
#include <stdlib.h>

void swap(int *a, int *b) {
int t = *a;
*a = *b;
*b = t;
}

int partition(int a[], int l, int r) {
int pivot = a[r];
int i = l - 1;

for(int j = l; j < r; j++) {
if(a[j] <= pivot) {
i++;
swap(&a[i], &a[j]);
}
}

swap(&a[i+1], &a[r]);
return i + 1;
}

int pivot_first(int l, int r) {
return l;
}

int pivot_last(int l, int r) {
return r;
}

int pivot_random(int l, int r) {
return l + rand() % (r - l + 1);
}
```

```
void quicksort(int a[], int l, int r, int (*pickPivot)(int,int)) {
if(l < r) {
int p = pickPivot(l, r);
swap(&a[p], &a[r]);
int pi = partition(a, l, r);
quicksort(a, l, pi - 1, pickPivot);
quicksort(a, pi + 1, r, pickPivot);
}
}

int main() {
int a[] = {157,110,147,122,111,149,151,141,123,112,117,133};
int n = sizeof(a)/sizeof(a[0]);

//quicksort(a, 0, n-1, pivot_first);
//quicksort(a, 0, n-1, pivot_last);
quicksort(a, 0, n-1, pivot_random);

for(int i = 0; i < n; i++)
printf("%d ", a[i]);

return 0;
}
```

OUTPUT:

```
C:\Users\mrgns\Desktop\daa\task5>gcc quicksortcases.c -o qsc

C:\Users\mrgns\Desktop\daa\task5>.\qsc
110 111 112 117 122 123 133 141 147 149 151 157
```

WORKING:

Quick Sort: Last element Pivot

Array is

[157, 110, 147, 122, 111, 149, 151, 141, 123, 112, 117, 133]

swap (157, 110]

[110, 157, 147, 122, 111, 149, 151, 141, 123, 112, 117, 133]

swap (157, 122]

[110, 122, 147, 157, 111, 149, 151, 141, 123, 112, 117, 133]

swap (147, 111)

[110, 122, 111, 157, 147, 149, 151, 141, 123, 112, 117, 133)

swap (157, 123)

[110, 122, 111, 123, 147, 149, 151, 141, 157, 112, 117, 133]

swap (147, 112)

[110, 122, 111, 123, 112, 149, 151, 141, 157, 147, 117, 133]

swap (149, 117)

[110, 122, 111, 123, 112, 117, 151, 141, 157, 147, 149, 133]

swap (151, 133)

[110, 122, 111, 123, 112, 117, 133, 141, 157, 149, 149, 151]

<133   >133

pivot = 117   pivot = 151

left array: (117)

swap (122, 111)

[110, 111, 122, 123, 112, 117]

swap (122, 112)

[110, 111, 112, 123, 122, 117]

swap (123, 117)

[110, 111, 112, 117, 122, 123] ⇒ sorted

Right array (151)

swap (157, 147)

[141, 147, 157, 149, 151]

swap (157, 149)

[141, 147, 149, 157, 151]

swap (157, 151)

[141, 147, 149, 151, 157] ⇒ sorted

merge them

Final sorted array:

[110, 111, 112, 117, 122, 123, 133, 141, 147, 149, 151, 157]

First element pivot (157)

swap it to last (157, 133)

[133, 110, 147, 122, 111, 149, 151, 141, 123, 112, 117, 157]

all elements less than 157 so no inner swap

final pivot stays

Pivot = 133

swap (133, 117)

[117, 110, 147, 122, 111, 149, 151, 141, 123, 112, 133, 157]

swap (147, 122)

[117, 110, 122, 147, 111, 149, 151, 141, 123, 112, 133, 157]

swap(147, 111)
[117, 110, 122, 111, 147, 149, 151, 141, 123, 112, 133, 157]

swap(147, 123)
[117, 110, 122, 111, 123, 149, 151, 141, 147, 112, 133, 157]

swap(149, 112)
[117, 110, 122, 111, 123, 112, 151, 141, 147, 149, 133, 157]

swap(151, 133)
[117, 110, 122, 111, 123, 112, 133, 141, 147, 149, 151, 157]

~~continue recursively~~ now pivot = 117 sorted

swap(117, 112)
[112, 110, 122, 111, 123, 117]

swap(122, 111)
[112, 110, 111, 122, 123, 117]

swap(122, 117)
[112, 110, 111, 117, 123, 122]

pivot = 112
swap(112, 110)
[110, 112, 111]

swap(111, 110)
[110, 111, 112]

Pivot = 123
swap(123, 122)
[122, 123] now all sorted so merge

[110, 111, 112, 117, 122, 123, 133, 141, 147, 149, 151, 157]

Random Pivot (111)

swap (111, 133)

[157, 110, 147, 122, 133, 149, 151, 141, 123, 112, 117, 111]

swap (157, 110)

[110, 157, 147, 122, 133, 149, 151, 141, 123, 112, 117, 111]

swap (157, 111)

[110, 111, 147, 122, 133, 149, 151, 141, 123, 112, 117, 157]

random pivot = 122 swap to 157 then

swap (147, 122)

[110, 111, 122, 157, 133, 149, 151, 141, 123, 147, 117, 122]

~~swap (157, 117)~~

~~[122, 112, 133, 149, 151, 141, 123, 147, 117, 157]~~

~~swap (133, 122)~~

~~[133, 112, 122, 149, 151, 141, 123, 147, 117, 157]~~

swap (157, 117)

[110, 111, 112, 117, 133, 149, 151, 141, 123, 147, 157, 122]

swap (133, 122)

[110, 111, 112, 117, 122, 149, 151, 141, 123, 147, 157, 133]

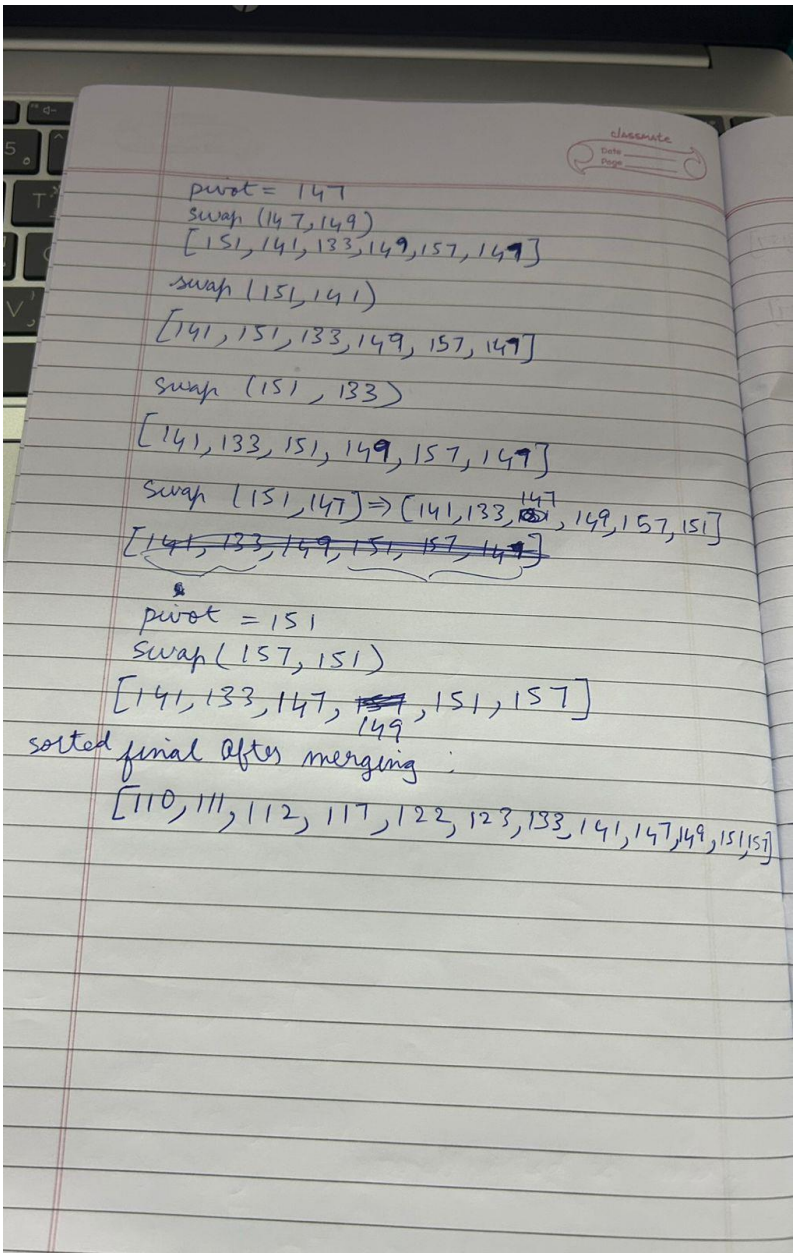
sorted

pivot = 123

swap it to last then

swap (149, 123)

[123, 151, 141, 133, 147, 157, 149]



pivot = 147
swap (147, 149)
[151, 141, 133, 149, 157, 147]
swap (151, 141)
[141, 151, 133, 149, 157, 147]
swap (151, 133)
[141, 133, 151, 149, 157, 147]
swap [151, 147] ⇒ [141, 133, 147, 149, 157, 151]
~~[141, 133, 149, 151, 157, 147]~~
pivot = 151
swap (157, 151)
[141, 133, 147, ~~157~~ 149, 151, 157]
sorted final after merging :
[110, 111, 112, 117, 122, 123, 133, 141, 147, 149, 151, 157]

TIME COMPLEXITY: O(nlogn) on average and best case, O(n^2) in the worst case due to unbalanced partitions.

SPACE COMPLEXITY: Due to recursion stack, worst case is O(n).

2. Quick Select

CODE:

```
#include <stdio.h>

int partition(int arr[], int low, int high) {
    int pivot = arr[high];
    int i = low;

    for (int j = low; j < high; j++) {
        if (arr[j] < pivot) {
            int temp = arr[i];
            arr[i] = arr[j];
            arr[j] = temp;
            i++;
        }
    }

    int temp = arr[i];
    arr[i] = arr[high];
    arr[high] = temp;

    return i;
}

int quickSelect(int arr[], int low, int high, int k) {
    if (low <= high) {
        int pivotIndex = partition(arr, low, high);

        if (pivotIndex == k)
            return arr[pivotIndex];
        else if (pivotIndex > k)
            return quickSelect(arr, low, pivotIndex - 1, k);
        else
            return quickSelect(arr, pivotIndex + 1, high, k);
    }
    return -1;
}

int main() {
    int arr[] = {157,110,147,122,111,149,151,141,123,112,117,133};
    int n = sizeof(arr) / sizeof(arr[0]);
    int k;

    printf("Enter k: ");
    scanf("%d", &k);

    int result = quickSelect(arr, 0, n - 1, k - 1);
    printf("k-th smallest element: %d\n", result);

    return 0;
}
```

OUTPUT:

```
C:\Users\mrgns\Desktop\daa>gcc quickselect.c -o qs

C:\Users\mrgns\Desktop\daa>.\qs
Enter k: 2
k-th smallest element: 111
```

WORKING:

Quick select

[157, 110, 147, 122, 111, 149, 151, 141, 123, 112, 117, 133]

k = 3

initial array :

[157, 110, 147, 122, 111, 149, 151, 141, 123, 112, 117, 133]

Choose pivot

pivot = 133 (last element)

Rearrange array elements <133 left of 133 & elements >133 to right of 133.

After partition,

[110, 122, 111, 123, 112, 117, 133, 141, 157, 151, 149]

pivot index = 6

pivot index > k [6 > 2] [index of k is 2]

As 6 > 2,

consider left subarray.

[110, 122, 111, 123, 112, 117]

new pivot = 117 (last element)

After partition,

[110, 111, 112, 117, 122, 123]

pivot index = 3

pivot index > k index [3 > 2]

now, [110, 111, 112]

new pivot = 112

After partition : [110, 111, 112]

pivot index = k [2 = 2]

k^{th} smallest number = 112.

TIME COMPLEXITY: O(n) on average and in the best case, $O(n^2)$ in the worst case due to highly unbalanced partitions.

SPACE COMPLEXITY: Due to recursion stack, average case is O(log n) and worst case is O(n).